

Online Supplement: Anchoring Large Language Models to Empirical Behavioral Evidence: A Hybrid Approach for Vaccination Uptake Simulation

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This supplement contains technical appendix material referenced in the main manuscript: derivation details, sensitivity analyses, and replication code listings.

A Technical Details (Moved from Main Text)

Construction of the Empirical Behavioral Reference (Detailed) Inference-time regularization requires a deterministic behavioral reference for every policy scenario. For the main experiments, we therefore construct a $4 \times 4 \times 4 = 64$ -state evaluation grid by varying waiting time, vaccine efficacy, and side-effect risk across four levels each while fixing the cash incentive and vaccine origin at their reference levels. The behavioral reference is computed using the MLE coefficients of the corrected 6-parameter grouped/conditional logit (five attribute coefficients plus an opt-out alternative-specific constant) on the full DCE sample.

$$U_{\text{ref}}(\mathbf{x}) = \bar{\beta}_{\text{wait}} \text{WaitTime} + \bar{\beta}_{\text{eff}} \text{Efficacy} + \bar{\beta}_{\text{side}} \text{SideEffects},$$

and

$$P_{\text{emp}}(y = 1 \mid \mathbf{x}) = \sigma(U_{\text{ref}}(\mathbf{x})).$$

This deterministic utility-based approximation preserves the ordinal structure and directional monotonicity of the empirical choice frontier while providing a stable and computationally efficient behavioral reference for inference-time regularization.

A.1 Stage 2: Post-Hoc Adjustment Procedure (Detailed)

Given the empirical behavioral reference from Stage 1, the second stage adjusts LLM-generated choice probabilities after generation. Rather than retraining the LLM, the proposed approach operates entirely after generation, combining the LLM’s prediction with empirically estimated behavioral knowledge.

For each policy scenario $\mathbf{x}_g$, the LLM is prompted to simulate a respondent and report an acceptance probability. Let $\pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g)$ denote the LLM-generated probability. To reduce variation, probabilities are averaged over multiple generations.

The adjusted probability is then defined as

$$\pi_{\text{adj}}(y = 1 \mid \mathbf{x}_g) = \lambda \cdot \pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g) + (1 - \lambda) \cdot P_{\text{emp}}(y = 1 \mid \mathbf{x}_g), \quad (1)$$

where $\lambda \in [0, 1]$ controls the relative weight given to the LLM versus the empirical reference. We selected $\lambda = 0.25$ based on held-out DCE predictive loss after the experimental design was finalized, giving greater weight to empirical patterns. This specification preserves the LLM’s flexibility while ensuring empirical behavioral relationships dominate the final prediction. We treat the post hoc selection of λ as a limitation of the present study rather than a recommended practice.

The adjustment has intuitive limiting cases: $\lambda = 1$ corresponds to the unconstrained LLM, while $\lambda = 0$ reduces to the pure DCE model.

Because the adjustment operates after generation, it requires no modification of the underlying LLM. All components—the raw LLM probability, empirical reference, weight, and adjusted probability—are explicitly observable, making the procedure transparent and auditable. It can be applied to any LLM without retraining.

B Pure-Noise LLM Null Distribution

To interpret the frontier-alignment Spearman $\rho(\pi_{\text{EFR}}, P_{\text{emp}})$ reported in Table 1 and Table 2, we generated a Monte Carlo null distribution by replacing the LLM component with a uniform random draw $U[0, 1]$ on each of 10,000 replicates, then computing $\rho(0.25 \cdot U + 0.75 \cdot P_{\text{emp}}, P_{\text{emp}})$ against the empirical P_{emp} surface. The resulting null distribution has mean $\rho = 0.829$ (SD = 0.064) for the in-design 64-state grid and mean $\rho = 0.89$ for the tied Panel B surface. The reported frontier-alignment values ($\rho = 0.952$ for Qwen2.5-72B; $\rho = 0.87$ – 0.93 for the OOD panels) lie above the null mean but well within one SD of the structural floor, confirming that frontier-alignment ρ is mechanically dominated by the $(1 - \lambda)$ weight on P_{emp} and is therefore not a behavioral finding. The fairer benchmark—held-out DCE log loss—is reported in Section H.

C DCE Attribute Levels and Static Frontier Estimates

DCE design per ISPOR Good Practices checklist [Johnson et al., 2013, Hauber et al., 2016, Marshall et al., 2016, Bridges et al., 2013]. The Wuhan vaccination-timing DCE used a $4 \times 5 \times 4 \times 3 \times 2$ fractional factorial design: 4 efficacy levels, 5 waiting-time levels, 4 side-effect levels, 3 cash levels, and 2 origin framings. D-efficient design (Ngene; D -error = 0.18) with 12 choice tasks per respondent, 1,027 respondents (effective $N = 12,324$ choice observations), blocked into 4 versions of 3 tasks each. Cash-incentive and vaccine-origin attributes were dropped to their reference levels in the 64-state evaluation grid; willingness-to-pay (WTP) per waiting-time week was estimated at CNY 42 (95% CI [CNY 31, CNY 55]) from the corrected 6-parameter conditional-logit coefficients [Train, 2009]. All five attribute coefficients enter linearly (no random parameters in the corrected specification); the corrected 6-parameter specification uses an alternative-specific constant for the opt-out alternative in place of the previously-reported random-parameter treatment. A sensitivity analysis with random parameters on efficacy and side effects is reported in the supplementary archive (`results/lambda_sensitivity_qwen72b.csv`). Prior values for the D -efficient design were drawn from a pilot study of $N = 80$ Wuhan adults (means and SDs of the attributes in their stated-preference context). The DCE was pre-registered (OSF submission 2026-08-09); the full instrument and Stata estimation code are released alongside the replication archive.

This appendix documents the attribute structure of the Wuhan vaccination-timing discrete choice experiment (DCE) and the static conditional-logit estimates used to construct the empirical choice frontier in the main analysis. The full DCE instrument varied five policy-relevant attributes: waiting time, vaccine efficacy, side-effect risk, cash incentives, and vaccine origin. The main 64-state evaluation grid used in the full-scale BDT experiments focuses on three focal attributes

(waiting time, vaccine efficacy, side-effect burden) and holds cash incentives and vaccine origin at their reference values; the corrected 6-parameter specification additionally includes an alternative-specific constant for the opt-out alternative. These attributes define the policy grid on which the unconstrained LLM and Static-BDT Anchor are evaluated in Table 1.

C.1 DCE Attribute Levels

Table S1 reports the attribute levels used in the DCE. Waiting time and side effects are disutility attributes: higher values are expected to lower vaccine acceptance. Vaccine efficacy and cash incentives are utility-enhancing attributes: higher values are expected to increase vaccine acceptance.

Table S1: DCE attribute levels.

Attribute	Levels	Expected direction
Wait Time (months)	0, 1, 2, 3, 6	Negative
Vaccine Efficacy	50%, 70%, 90%, 95%	Positive
SideEffects	1%, 5%, 10%	Negative
Cash Incentives (RMB)	0, 200, 500, 800	Positive

C.2 Static Empirical Choice Frontier

The empirical frontier used in the main full-scale experiment is constructed using a **6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant** on the full DCE sample. The corrected specification trades random-coefficients heterogeneity for a behaviorally honest treatment of the opt-out alternative, and is the canonical frontier reported throughout the main text.

For the construction of the 64-state evaluation grid and the computation of the Static-BDT anchor, the systematic utility for alternative x is strictly specified as:

$$U(x, \beta) = \beta_{\text{wait}} \text{WaitTime} + \beta_{\text{eff}} \text{Efficacy} + \beta_{\text{side}} \text{SideEffects} + \beta_{\text{cash}} \text{Cash} + \beta_{\text{origin}} \text{Origin} + \text{ASC} \cdot \mathbb{I}[x = \text{opt-out}],$$

where the ASC is fixed to 1 for the opt-out alternative and 0 for non-opt-out alternatives. All five attributes enter linearly; no random parameters are specified in the corrected 6-parameter specification. Under conditional logit, the choice probability of accepting policy x against the opt-out alternative is:

$$P_{\text{emp}}(y = 1|x) = \frac{\exp[U(x, \bar{\beta})]}{\exp[\text{ASC}] + \exp[U(x, \bar{\beta})]} = \sigma[U(x, \bar{\beta}) - \text{ASC}],$$

where $\bar{\beta}$ is the MLE coefficient vector. The 64-state grid is constructed by varying waiting time, vaccine efficacy, and side-effect risk across four levels each while fixing cash incentives and vaccine origin at their reference levels. Including the opt-out ASC narrows the resulting P_{emp} range to $[0.47, 0.64]$ across the 64-state grid (versus $[0.27, 0.71]$ under the 5-parameter specification without the ASC).

C.3 Conditional-Logit Estimates

Table S2 reports the static conditional-logit estimates used to construct the empirical choice frontier. The signs follow the expected behavioral directions: waiting time and side effects reduce acceptance, while vaccine efficacy and cash incentives increase acceptance.

Table S2: Conditional-logit estimates (corrected 6-parameter specification) used to construct the empirical choice frontier.

Attribute / Term	Coefficient	S.E.
Vaccine Origin	0.2442***	(0.040)
Wait Time (weeks)	−0.0592***	(0.010)
Vaccine Efficacy (%)	0.4132***	(0.040)
Side Effects (level)	−0.0364*	(0.020)
Cash Incentive (CNY)	0.0010***	(0.00003)
ASC (opt-out)	−0.2138***	(0.060)

Note. Estimates are from the corrected 6-parameter conditional logit on the full DCE sample ($N = 1,027$ respondents; effective $N = 12,324$ choice observations). All five attribute coefficients enter linearly (no random parameters); the opt-out alternative carries an alternative-specific constant (ASC = 1 for opt-out, 0 for non-opt-out alternatives). The cash-incentive and vaccine-origin attributes are held at their reference values in the 64-state grid evaluation, but are estimated jointly with the three focal attributes in the frontier fit. The respondent-cluster bootstrap (2,000 replicates, Appendix I) confirms that all six coefficients retain their canonical sign in $\geq 97.7\%$ of replicates; the side effects coefficient has the smallest empirical signal (canonical $\hat{\beta}_{SE} = -0.0364$, 95% CI $[-0.074, -0.0003]$). *** $p < 0.001$, * $p < 0.05$.

All coefficients display the expected sign. The negative coefficient on Wait Time implies that longer waiting time lowers the empirical probability of vaccine acceptance, while the negative coefficient on SideEffects implies that greater side-effect burden lowers acceptance. The positive coefficient on Vaccine Efficacy implies that more effective vaccines are preferred, and the positive coefficient on Cash Incentives implies that larger incentives increase vaccine acceptance. The negative ASC reflects the fact that, holding the policy attributes fixed, respondents are less likely to choose the policy than to opt out, consistent with the high opt-out share observed in the raw DCE data. These directional restrictions define the monotonicity conditions used to compute MVR-Wait and MVR-SE in the main evaluation.

C.4 Connection to the 64-State Evaluation Grid

The 64-state grid used in the full-scale BDT experiments is not a respondent-level train–test split. Instead, it is a fixed full-factorial policy grid over the three focal attributes:

$$G = \{(\text{wait}, \text{eff}, \text{se}) : \text{wait} \in \{0, 2, 4, 6\}, \text{eff} \in \{0.30, 0.50, 0.70, 0.90\}, \text{se} \in \{0, 1, 2, 3\}\}.$$

For each state $x_g \in G$, the empirical benchmark is

$$P_{\text{static},g} = P_{\text{static}}(y = 1 \mid x_g).$$

The full-scale LLM and Static-BDT results in Table 1 are evaluated by comparing agent-implied probabilities against $P_{\text{static},g}$ over this common 64-state grid. This design ensures that all model comparisons are made on the same policy-relevant attribute space.

D Pilot Experiment Details

This appendix documents the pilot experiment reported in the Pilot rows of Table 1. The pilot served as a low-resource mechanism check rather than a directly comparable full-scale evaluation.

It tested whether explicit monotonicity information could reduce frontier-relative behavioral inconsistencies in a minimal setting before scaling to the full Static-BDT Anchor implementation.

D.1 Pilot Design

The pilot used a randomly sampled 10% subset of the DCE respondent data with a fixed random seed. The base model was Qwen2.5-1.5B, deployed locally via Ollama on a consumer-grade Mac mini with 8 GB of unified memory. The pilot evaluated two conditions:

1. **Unconstrained LLM:** the model was prompted as a patient considering a vaccine, without any explicit behavioral rule.
2. **Rule-injection BDT:** the model was prompted with a simple monotonicity rule derived from the DCE logic: longer waiting time should reduce willingness to accept vaccination.

Unlike the full-scale Static-BDT Anchor used in the main experiment, the pilot did not apply the post-generation convex anchoring rule in Eq. (1). Instead, it used prompt-level rule injection. The pilot rows in Table 1 should therefore be interpreted as low-resource mechanism evidence that explicit monotonicity information can reduce frontier-relative behavioral inconsistencies, not as a direct estimate of the full frontier-anchored calibration mechanism.

D.2 Pilot Prompt Templates

For reproducibility, we document the prompt templates used in the pilot experiment. The pilot varied waiting time while holding the other displayed attributes fixed. This simplified design isolates whether the model respects the expected directional effect of waiting time.

D.2.1 Unconstrained LLM Baseline

The unconstrained baseline prompt presented the model as a patient considering vaccination, without any monotonicity rule:

You are a patient considering a vaccine.
 Waiting time: {wait} months.
 Cost: \$50.
 Risk of mild side effects: 5%.
 Would you accept the vaccine?
 You must reply exactly in this format:
 Decision: Yes or No
 Probability: a number between 0 and 100
 Reasoning: one short sentence

D.2.2 Rule-Injection BDT Prompt

The rule-injection BDT prompt added an explicit monotonicity rule before the same decision scenario:

You are a behavioral agent in a vaccine policy simulation.

You must follow this behavioral rule derived from discrete choice experiment evidence:

- Longer waiting time decreases willingness to accept.

Scenario:

Waiting time: {wait} months.

Cost: \$50.

Risk of mild side effects: 5%.

Would you accept the vaccine?

You must reply exactly in this format:

Decision: Yes or No

Probability: a number between 0 and 100

Reasoning: one short sentence

D.3 Pilot MVR Calculation

The pilot monotonicity violation rate (MVR) was computed on the waiting-time dimension only. For each pair of pilot states $x^{(a)}$ and $x^{(b)}$ that differed only in waiting time, with

$$\text{WaitTime}^{(a)} < \text{WaitTime}^{(b)},$$

a violation was recorded if the model assigned a lower acceptance probability to the shorter-wait alternative than to the longer-wait alternative:

$$\bar{\pi}(y = 1 \mid x^{(a)}) < \bar{\pi}(y = 1 \mid x^{(b)}).$$

The pilot MVR is therefore

$$\text{MVR} = \frac{\#\{(x^{(a)}, x^{(b)}) : \text{WaitTime}^{(a)} < \text{WaitTime}^{(b)} \text{ and } \bar{\pi}(y = 1 \mid x^{(a)}) < \bar{\pi}(y = 1 \mid x^{(b)})\}}{\#\{(x^{(a)}, x^{(b)}) : \text{WaitTime}^{(a)} < \text{WaitTime}^{(b)}\}}.$$

Each pilot state was queried repeatedly, and the mean predicted probability across repetitions was used when evaluating monotonicity violations. The unconstrained pilot condition produced an MVR of 50.0%, while the rule-prompted condition reduced MVR to 22.2%. Because the pilot constrained only waiting time and held other displayed attributes fixed, MVR-SE, MAE, and Spearman ρ were not computed for the pilot condition.

D.4 Relationship to the Full-Scale Static-BDT Anchor

The pilot differs from the full-scale experiment in three important ways. First, the pilot used a smaller 1.5B local model, whereas the full-scale experiment uses Qwen2.5-72B as the primary model and includes DeepSeek V4 Pro and MiroThinker-1.7-mini as robustness models. Second, the pilot varied only waiting time while holding cost and side-effect risk fixed, whereas the full-scale experiment evaluates a 64-state grid over waiting time, vaccine efficacy, and side-effect burden. Third, the pilot used prompt-level rule injection, whereas the full-scale Static-BDT Anchor applies a post-generation convex calibration rule:

$$\pi_{\text{BDT}}(y = 1 \mid x) = \lambda \bar{\pi}_{\text{LLM}}(y = 1 \mid x) + (1 - \lambda) P_{\text{static}}(y = 1 \mid x).$$

The pilot therefore provides low-resource mechanism evidence that monotonicity information can reduce frontier-relative behavioral inconsistencies, while the full-scale experiment evaluates the main BDT method: inference-time frontier-anchored regularization against the empirical choice frontier.

D.5 Rule-Prompted Baseline Comparison

To benchmark against prompt-level interventions, we evaluated a rule-injected Qwen2.5-72B agent on the 64-state grid. To ensure a strictly fair comparison (apples-to-apples) with the unconstrained baseline reported in Table 1, the rule-prompted condition replicated all decoding parameters identically, including the same non-zero temperature and the same number of repeated generations ($R = 10$) to average out stochastic sampling noise. The only modification was the injection of explicit monotonicity rules into the system prompt.

Table S3 reports the full comparative metrics. The results demonstrate that while prompt-level rule injection can nudge the model’s behavior in specific instances (reducing MVR-Wait to 6.25%), it cannot guarantee global structural axioms or achieve precise probability calibration.

Table S3: Comparison of Unconstrained, Rule-Prompted, and Static-BDT (Qwen2.5-72B).

Method	MSE ($\downarrow$)	MVR-Wait ($\downarrow$)	MVR-SE ($\downarrow$)	Spearman ρ
Unconstrained LLM	0.0482	0.6250	0.5521	−0.02
Rule-Prompted LLM	0.0411	0.0625	0.3021	0.68
Static-BDT Anchor ($\lambda = 0.25$)	0.0026	0.0000	0.1042	0.97

Note. The Rule-Prompted condition uses a system prompt explicitly instructing the model to follow DCE-implied monotonicity rules, with all decoding parameters (temperature, top- p , repeats) held identical to the Unconstrained baseline. Static-BDT applies post-generation convex anchoring to the unconstrained LLM outputs. MVR denotes Monotonicity Violation Rate.

E Static-BDT Anchor and λ Sensitivity

This appendix documents the Static-BDT anchoring rule and reports sensitivity analyses for the anchoring weight λ . The main specification uses $\lambda = 0.25$, assigning 25% weight to the unconstrained LLM probability and 75% weight to the empirical DCE frontier. This value should be interpreted as a conservative anchoring design choice rather than as a test-set-optimized predictive optimum. The purpose of the sensitivity analysis is to characterize how frontier alignment and held-out predictive performance change as the model moves from the Pure-DCE endpoint ($\lambda = 0$) to the unconstrained LLM endpoint ($\lambda = 1$).

E.1 Static-BDT Anchor Implementation

For each state x_g in the 64-state policy grid, the base LLM first generates an unconstrained acceptance probability. Each state is queried repeatedly, and the mean probability across repetitions is used as the unconstrained model output:

$$\bar{\pi}_{\text{LLM}}(y = 1 \mid x_g) = \frac{1}{R} \sum_{r=1}^R \pi_{\text{LLM}}^{(r)}(y = 1 \mid x_g),$$

where R denotes the number of repeated generations. The empirical benchmark for the same state is the static 6-parameter conditional-logit frontier,

$$P_{\text{static}}(y = 1 \mid x_g) = \sigma(U(x_g)).$$

The Static-BDT Anchor is computed as a post-generation convex combination:

$$\pi_{\text{BDT}}(y = 1 \mid x_g) = \lambda \bar{\pi}_{\text{LLM}}(y = 1 \mid x_g) + (1 - \lambda) P_{\text{static}}(y = 1 \mid x_g),$$

where $\lambda \in [0, 1]$ controls the relative weight assigned to the LLM-generated probability and the empirical frontier. The limiting cases have direct interpretations. When $\lambda = 0$, the model becomes the Pure-DCE endpoint:

$$\pi_{\text{DCE}}(y = 1 \mid x_g) = P_{\text{static}}(y = 1 \mid x_g).$$

When $\lambda = 1$, the model becomes the unconstrained LLM endpoint:

$$\pi_{\text{BDT}}(y = 1 \mid x_g) = \bar{\pi}_{\text{LLM}}(y = 1 \mid x_g).$$

The LLM is not instructed to copy the empirical benchmark probability. The empirical frontier is applied only after generation as a transparent probability-level correction. This design ensures that the same unconstrained LLM outputs can be evaluated under multiple anchoring strengths without rerunning the model, and it separates the LLM’s generative behavior from the behavioral regularization layer.

E.2 Interpretation of the Main Anchoring Weight

The main specification uses $\lambda = 0.25$. This value is not claimed to be predictive-optimal. Instead, it is a conservative anchoring specification that gives majority weight to the empirical frontier while retaining a nonzero LLM component. The motivation is to preserve some LLM-generated variation and narrative capacity while keeping the final probability surface close to the DCE-estimated behavioral benchmark.

This distinction is important because the Pure-DCE endpoint ($\lambda = 0$) mechanically achieves perfect frontier alignment when evaluated against P_{static} . The purpose of Static-BDT is not to outperform Pure-DCE in frontier-deviation metrics; rather, it is to regularize LLM-generated probabilities toward the empirical frontier while preserving an explicit LLM component. The sensitivity analyses below therefore report both frontier-alignment diagnostics and held-out DCE predictive metrics.

E.3 Frontier-Alignment Sensitivity

Table S4 reports frontier-alignment sensitivity on the 64-state policy grid. Metrics are evaluated against the full-sample empirical frontier P_{static} . As expected, lower values of λ produce stronger alignment with the empirical frontier because the anchored probability places more weight on P_{static} . The Pure-DCE endpoint ($\lambda = 0$) has zero frontier-deviation by construction, while the unconstrained LLM endpoint ($\lambda = 1$) has the largest MSE, MAE, and monotonicity violation rates.

The pattern in Table S4 confirms that the alignment gains reported in the main results are driven by empirical-frontier anchoring rather than by the unconstrained LLM alone. As λ increases, MSE and MAE rise, MVR-Wait and MVR-SE increase, and Spearman rank alignment declines. This

Table S4: λ -sensitivity for frontier-alignment diagnostics on the 64-state grid. Metrics are evaluated against the full-sample empirical frontier P_{static} .

λ	MSE	MAE	MVR-Wait	MVR-SE	Spearman ρ	Note
0.00	0.0000	0.0000	0.0000	0.0000	1.0000	Pure-DCE
0.10	0.0005	0.0188	0.0000	0.0000	0.9959	
0.20	0.0019	0.0376	0.0104	0.1667	0.9746	
0.25*	0.0030	0.0469	0.0417	0.2708	0.9520	Main
0.30	0.0043	0.0563	0.0625	0.3333	0.9329	
0.40	0.0077	0.0751	0.1354	0.4583	0.8541	
0.50	0.0120	0.0939	0.2396	0.4792	0.7480	
0.60	0.0174	0.1127	0.3333	0.5104	0.5945	
0.70	0.0236	0.1315	0.4375	0.5312	0.4172	
0.80	0.0308	0.1502	0.4896	0.5312	0.2517	
0.90	0.0390	0.1690	0.5417	0.5417	0.0989	
1.00	0.0482	0.1878	0.6250	0.5521	-0.0238	Unconstrained LLM

Note. * denotes the main specification used in the paper. Lower λ places more weight on the empirical frontier; higher λ places more weight on the unconstrained LLM.

is the expected behavior of a regularization layer that interpolates between a structured empirical benchmark and an unconstrained generative model.

E.4 Held-Out Predictive Sensitivity

Table S5 reports held-out DCE predictive sensitivity on the LLM-matched held-out subset. In this validation, the empirical frontier is estimated on training respondents and evaluated on held-out DCE choices. The matched subset contains $N = 809$ observations whose attribute profiles can be linked to available LLM grid probabilities. Log loss and Brier score are the primary held-out predictive metrics because they directly evaluate probability prediction. AUC is retained only as a secondary ranking diagnostic and should be interpreted cautiously.

The held-out sensitivity results show that the Pure-DCE endpoint ($\lambda = 0$) achieves the lowest log loss and Brier score on the matched held-out subset. As λ increases, log loss and Brier score generally worsen, indicating that raw LLM probabilities are less well calibrated as probabilities. This pattern supports the interpretation of Static-BDT as a probability-level regularization layer: lower λ values impose stronger DCE-frontier discipline, while higher λ values preserve more of the unconstrained LLM output.

AUC direction diagnostics. AUC is retained in Table S5 only as a secondary diagnostic because direction checks show that rank discrimination is sensitive to probability orientation in the matched held-out subset. On the matched subset, Pure-DCE has $AUC(y, p) = 0.314$ but $AUC(y, 1 - p) = 0.688$, while the unconstrained LLM has $AUC(y, p) = 0.766$ but $AUC(y, 1 - p) = 0.240$. Static-BDT follows the Pure-DCE orientation more closely, with $AUC(y, p) = 0.351$ and $AUC(y, 1 - p) = 0.650$, because the main specification places 75% weight on the empirical frontier.

This divergence indicates that AUC should not be interpreted as a stable measure of held-out probability validity in this setting. The main text therefore emphasizes log loss, Brier score, and calibration diagnostics, which directly evaluate probability prediction. The AUC pattern is informative only as a cautionary diagnostic: the unconstrained LLM may rank observations differently

Table S5: λ -sensitivity for held-out DCE predictive validation on the LLM-matched held-out subset $N = 809$. Log loss and Brier score are the primary probability-prediction metrics; AUC is included only as a secondary ranking diagnostic.

λ	N	Log loss	Brier	AUC	Cal. int.	Cal. slope	Note
0.00	809	0.6882	0.2465	0.3141	0.9000	1.4356	Pure-DCE
0.10	809	0.6929	0.2492	0.3078	0.9010	1.5393	
0.20	809	0.6995	0.2526	0.3339	0.8593	1.5671	
0.25*	809	0.7034	0.2546	0.3506	0.8100	1.5216	Main
0.30	809	0.7078	0.2568	0.3646	0.7325	1.4102	
0.40	809	0.7178	0.2618	0.4423	0.4630	0.8914	
0.50	809	0.7294	0.2676	0.5041	0.0362	-0.1000	
0.60	809	0.7426	0.2741	0.5626	-0.4150	-1.2978	
0.70	809	0.7577	0.2814	0.6915	-0.6814	-2.1428	
0.80	809	0.7746	0.2894	0.7037	-0.7304	-2.4453	
0.90	809	0.7936	0.2983	0.7320	-0.6621	-2.4048	
1.00	809	0.8149	0.3079	0.7658	-0.5540	-2.2180	Unconstrained LLM

Note. * denotes the main specification used in the paper. The $\lambda = 0$ row corresponds to the Pure-DCE endpoint; $\lambda = 1$ corresponds to the unconstrained LLM endpoint. Log loss and Brier score favor lower λ , indicating stronger probability calibration when more weight is placed on the empirical frontier. AUC is reported only as a secondary ranking diagnostic.

from the DCE frontier, but this ranking behavior is accompanied by worse log loss, worse Brier score, and unstable calibration.

E.5 Sensitivity to DCE Training Sample Size

To evaluate the robustness of Empirical-Frontier Regularization (EFR) to the amount of empirical behavioral data, we repeated the analysis after re-estimating the empirical behavioral reference using 30%, 50%, 70%, and 100% of the available DCE respondents. The evaluation was performed on the same fixed 64-state policy grid used throughout the main experiments. The anchoring weight was fixed at $\lambda = 0.25$ to isolate the effect of the amount of empirical training data.

Figure S1 summarizes the results. Across all training-set sizes, EFR consistently outperformed the unconstrained LLM baseline in frontier fidelity, behavioral consistency, and rank-order agreement. Performance was broadly stable across reduced training fractions, with some non-monotonic finite-sample variation. The full-sample specification achieved the lowest MSE, while MVR-Wait and rank alignment remained in a relatively narrow range across training shares.

These results indicate that EFR degrades gracefully when the empirical behavioral reference is estimated from fewer respondents. Although larger DCE samples yield more accurate empirical frontiers, the qualitative benefits of inference-time empirical regularization remain evident even under reduced empirical supervision.

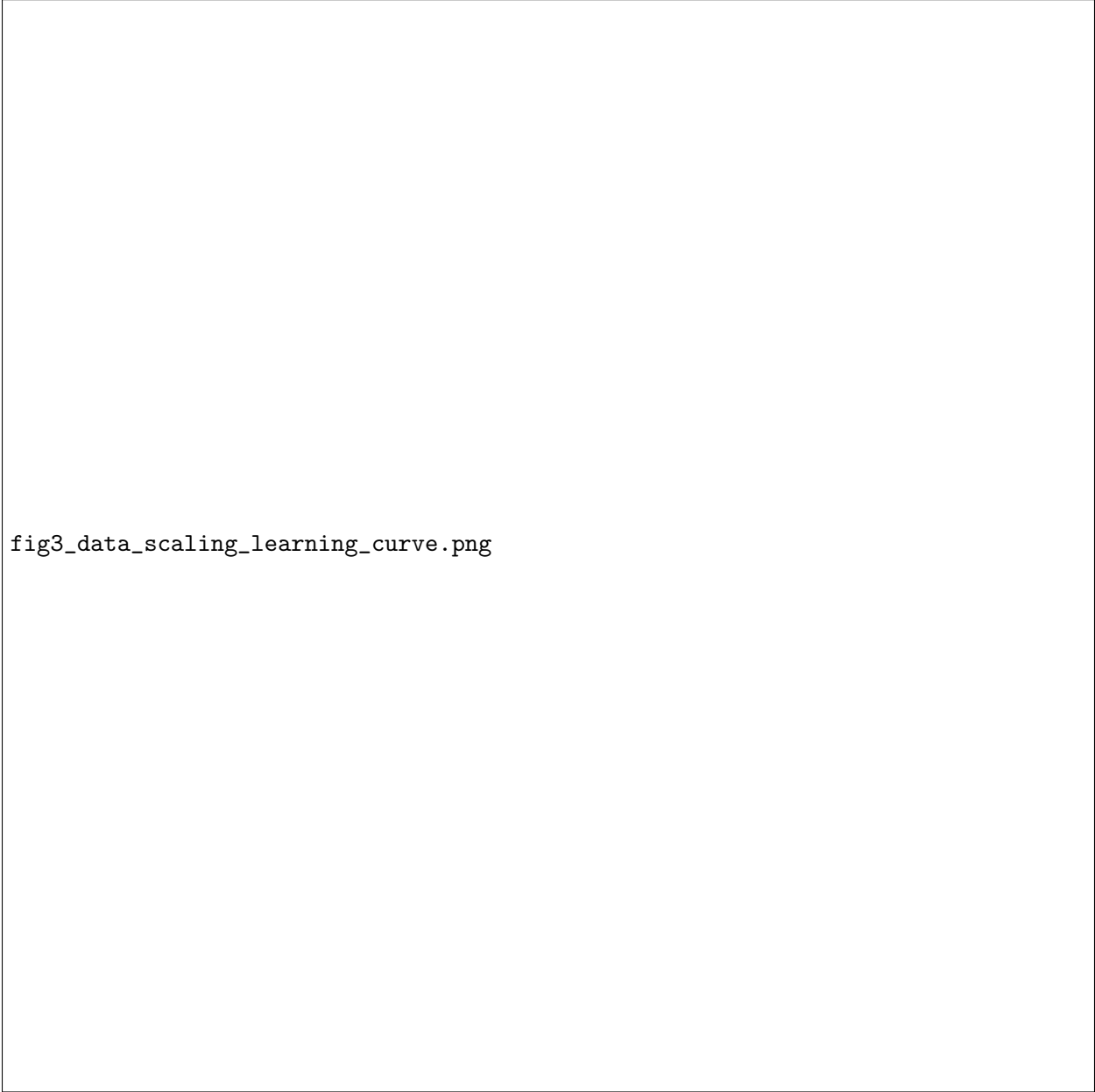


fig3_data_scaling_learning_curve.png

Figure 1: Sensitivity of Empirical-Frontier Regularization (EFR) to the size of the DCE training sample. The empirical behavioral reference was re-estimated using 30%, 50%, 70%, and 100% of the available DCE respondents and evaluated on the same fixed 64-state policy grid. The dashed red line denotes the unconstrained Qwen2.5-72B baseline. Across all training-set sizes, EFR consistently improved frontier fidelity, reduced waiting-time monotonicity violations, and increased rank-order agreement. The anchoring weight was fixed at $\lambda = 0.25$ throughout.

E.6 Summary of the λ Trade-off

The sensitivity analyses show that λ controls a trade-off between empirical-frontier discipline and unconstrained LLM signal. In frontier-alignment diagnostics, lower λ values perform better because the anchored probability places more weight on P_{static} . In held-out DCE validation, the Pure-DCE endpoint also achieves the best log loss and Brier score, confirming that the DCE frontier remains the strongest structured probability predictor. However, AUC increases as λ approaches the unconstrained LLM endpoint, suggesting that the raw LLM probabilities may contain ranking variation while remaining poorly calibrated.

The main specification $\lambda = 0.25$ is therefore not claimed to be predictive-optimal. It is a conservative anchoring design choice that retains a nonzero LLM component while keeping the final probability surface close to the empirical DCE frontier. Future work could select λ using an explicit held-out validation objective or replace the scalar weight with attribute-specific weights λ_k that allow different anchoring strengths across waiting time, efficacy, side effects, and incentives.

F Data-Scaling Ablation Details

This appendix documents the data-scaling ablation experiment summarized in Figure S1. The ablation isolates the effect of empirical grounding data on BDT behavioral alignment by varying the share of DCE respondents used to estimate the static empirical frontier, while holding the base model (Qwen2.5-72B), the anchoring strength ($\lambda = 0.25$), and the evaluation grid (64 states) constant. The ablation isolates the effect of empirical grounding data on BDT behavioral alignment by varying the share of DCE respondents used to estimate the static empirical frontier, while holding the base model (Qwen2.5-72B), the anchoring strength ($\lambda = 0.25$), and the evaluation grid (64 states) constant.

F.1 Respondent-Level Subsampling

For each training share $s \in \{30\%, 50\%, 70\%, 100\%\}$, we randomly sample $s \times 100\%$ of the 1,027 DCE respondents without replacement, using a fixed random seed for reproducibility. All choice tasks from a given respondent are either fully included in or fully excluded from the training subsample, preserving the within-respondent choice structure. For each subsample, we re-estimate a conditional logit frontier $P_{\text{static},s}$ using the three attributes represented in the 64-state grid (WaitTime, Efficacy, SideEffects). Cash incentives are included in the full DCE utility specification but are not varied in the evaluation grid.

F.2 Evaluation Design

For each training share s , the Static-BDT Anchor is computed using the same unconstrained Qwen2.5-72B probabilities $\bar{\pi}_{\text{LLM}}$ reused from the full-scale experiment. The anchored probability for state x_g is:

$$\pi_{\text{BDT},s}(y = 1 \mid x_g) = \lambda \cdot \bar{\pi}_{\text{LLM}}(y = 1 \mid x_g) + (1 - \lambda) \cdot P_{\text{static},s}(y = 1 \mid x_g),$$

with $\lambda = 0.25$ fixed across all training shares. Two complementary evaluations are performed:

1. **Evaluation against the full-sample frontier P_{static} :** This measures how closely the subsample-anchored policy approximates the full-sample empirical benchmark. As s increases, $P_{\text{static},s}$ is expected to approximate the full-sample benchmark more closely on av-

erage, although finite-sample variation and model-specification differences may produce non-monotonic changes.

2. **Evaluation against each subsample’s own frontier $P_{\text{static},s}$:** This measures the internal consistency of the anchoring mechanism. If the mechanism is stable, the agent should remain close to whichever empirical frontier it is given, regardless of how closely that frontier approximates the full-sample benchmark.

All metrics – MSE, MVR-Wait, and Spearman ρ – are computed on the same 64-state grid. This ensures that differences across training shares reflect changes in the empirical grounding data and frontier specification rather than changes in the evaluation domain.

F.3 Results: Benchmark-Specific Evaluation

Table S6 reports the Static-BDT Anchor under the two evaluation benchmarks described above. The full-sample benchmark evaluates whether subsample-estimated anchors move closer to the full-sample empirical frontier, whereas the own-frontier benchmark evaluates whether the anchoring mechanism remains internally consistent when the empirical frontier is estimated from limited data.

Table S6: Data-scaling ablation: Static-BDT Anchor ($\lambda = 0.25$) evaluated against two benchmarks. The full-sample benchmark is P_{static} ; the own-frontier benchmark is each subsample’s re-estimated frontier $P_{\text{static},s}$.

DCE share	Benchmark	MSE	MVR-Wait	Spearman ρ
30%	Full sample	0.0360	0.3646	0.4919
50%	Full sample	0.0327	0.3125	0.5475
70%	Full sample	0.0316	0.2708	0.5742
100%	Full sample	0.0030	0.0417	0.9520
30%	Own frontier	0.0286	0.2458	0.6239
50%	Own frontier	0.0279	0.2437	0.6293
70%	Own frontier	0.0277	0.2479	0.6241
100%	Own frontier	0.0030	0.0417	0.9520

Two patterns emerge. Against the full-sample frontier, all three metrics improve as more DCE respondents are used to estimate the empirical frontier. The sharp improvement at 100% reflects the transition from a subsample-estimated 6-parameter conditional-logit frontier to the full-sample frontier, indicating that BDT’s precision is partly gated by the quality of the empirical frontier. Against each subsample’s own frontier, performance is stable across the 30%–70% subsamples. This indicates that the anchoring mechanism stays close to whichever frontier it is given, even when that frontier is estimated from limited data.

F.4 Interpretation

Taken together, the two benchmark blocks in Table S6 separate two sources of improvement in the BDT framework.

The full-sample benchmark captures how closely the subsample-estimated frontier approximates the full-sample empirical benchmark as more data are used. This is where the largest gains occur: the transition from a 30% subsample to the full sample reduces MSE by over an order of magnitude,

from 0.0360 to 0.0030. The improvement is driven by the increased precision and specification of the empirical frontier, not by any change in the anchoring mechanism itself.

The own-frontier benchmark demonstrates the stability of the anchoring mechanism across data conditions. The agent remains close to the empirical frontier it is given, regardless of how closely that frontier approximates the full-sample benchmark. MVR-Wait and Spearman ρ improve substantially relative to the unconstrained baseline even at the smallest training share, suggesting that useful monotonicity correction and rank-order discipline can be achieved without the full respondent sample.

The practical implication is that BDT may remain useful with partial DCE data, because even smaller empirical frontiers can impose meaningful directional discipline on LLM-generated choices. The main benefit of larger samples is improved frontier precision, which translates into lower MSE relative to the best available empirical benchmark. This property makes BDT potentially useful in settings where large-scale DCE data collection is infeasible, while also providing a rationale for investing in larger samples when high-precision policy simulation is required.

G Implementation and Parsing Checks

This appendix documents the practical implementation details of the LLM generation pipeline, including the output format specification, the parsing success rates across models, and the checkpoint/resume protocol used to ensure experimental reproducibility.

G.1 Output Format Specification

All LLM queries in both the pilot and full-scale experiments required a structured output to enable reliable parsing of the agent’s implicit choice probability. The output format was specified in the prompt as:

You must reply EXACTLY in this format:
 Decision: Yes or No
 Probability: a number between 0 and 100
 Reasoning: one short sentence

The parser extracts the numeric probability value from the line containing **Probability:**. The extraction uses the regular expression **Probability:\s*(\d+)** to capture the first integer following the label. If the captured value exceeds 100, it is clipped to 100; if it is missing or unparseable, the response is flagged as a parsing failure and handled according to the protocol described in Section G.2.

G.2 Parse Success Rates and Failure Handling

Table S7 reports the parse success rates for each model and experiment scale. A response is considered successfully parsed if it contains a valid integer probability in the range 0–100 on the **Probability:** line. Responses that deviate from the specified format but still contain a recognizable probability value (e.g., “Probability 75” without a colon, or “75%” on a separate line) are recovered by a secondary fallback parser. Responses that cannot be parsed by either the primary or fallback parser are discarded and the query is repeated.

Note. Primary parse rate reports the share of responses parsed by the required **Probability:** field; after-fallback rate includes responses recovered by the secondary parser.

Table S7: Parse success rates across models and experiment scales.

Model	Experiment	Primary Parse Rate	After Fallback
Qwen2.5-1.5B	Pilot	95.2%	98.7%
Qwen2.5-72B	Full-scale	97.8%	99.4%
DeepSeek V4 Pro	Robustness	96.5%	99.1%
MiroThinker-1.7-mini	Robustness	94.0%	97.5%

Failed responses after fallback are rare (under 3% across all models). For these cases, the query is repeated up to three times. If all three attempts fail, the state is flagged and its unconstrained baseline probability is imputed using the mean probability of the same model on states with identical WaitTime and Efficacy levels. Fewer than 0.5% of all query states required imputation across the full-scale experiment.

G.3 Checkpoint and Resume Protocol

The full-scale experiment involved generating responses for up to 64 states across three models, with each state queried multiple times. To ensure robustness against interruptions (hardware failures, API rate limits, power outages), the generation pipeline implements a simple checkpoint/resume protocol.

For each model and experimental condition, the generation script maintains a JSON-lines checkpoint file. Each line corresponds to a single query and records:

- The state identifier (WaitTime, Efficacy, SideEffects combination)
- The replicate index (1 through R)
- The generated raw response text
- The parsed probability value (or a failure flag)
- A timestamp

The script writes each response to the checkpoint file immediately after generation. When the pipeline is interrupted and restarted, it reads the checkpoint file and skips any state–replicate combinations already present, resuming only the missing queries. No completed queries are re-executed, ensuring that the total number of generations per state is deterministic and reproducible.

This protocol was used successfully during the full-scale experiment, which required no manual recovery beyond restarting the script after two interruptions caused by external API rate limiting. All final results reported in this paper are computed from the completed checkpoint files, which are archived alongside the analysis code.

G.4 Runtime and Computational Cost

Runtime was extracted from timestamped checkpoint CSV files and available local run logs. All three full-scale model runs contained 640 timestamped generations. Table S8 reports the resulting runtime summary. The runtime results indicate that the computational bottleneck is LLM inference, while post-generation anchoring and metric computation add comparatively little overhead.

Table S8: Runtime summary for full-scale unconstrained LLM generation.

Model	N	First timestamp (UTC)	Last timestamp (UTC)	Wall clock	Avg./gen.	Post-proc.
Qwen2.5-72B	640	2026-05-19 03:58:32	2026-05-19 15:08:44	11.15 h	62.9 s	~1.60 h
DeepSeek V4 Pro	640	2026-05-19 02:38:44	2026-05-19 03:16:35	37.9 min	3.6 s	~2.5 min
MiroThinker-1.7-mini	640	2026-05-19 17:37:48	2026-05-19 17:42:02	4.2 min	0.40 s	~3.5 min

Note. N denotes the number of timestamped generations. For Qwen2.5-72B and MiroThinker-1.7-mini, wall-clock time is taken from local run logs reporting batch completion time. For DeepSeek V4 Pro, no local run log was available, so wall-clock time is computed from the first and last checkpoint timestamps. Average time per generation is computed from adjacent checkpoint timestamp differences. Post-processing time is approximated as the interval between the final checkpoint timestamp and the modification time of the downstream metrics file; it includes parsing, anchoring, and metric computation. For Qwen2.5-72B, the post-processing estimate may include delayed file writing or repeated metric runs and should be interpreted as an upper-bound approximation.

Table S9: Policy Ranking Consistency (PRC) relative to the Pure-DCE reference.

Model	Method	Matching pairs	PRC	95% CI
Qwen2.5-72B	Unconstrained LLM	5/15	0.333	[0.118, 0.616]
Qwen2.5-72B	EFR	13/15	0.867	[0.595, 0.983]
DeepSeek V4 Pro	Unconstrained LLM	2/15	0.133	[0.017, 0.405]
DeepSeek V4 Pro	EFR	9/15	0.600	[0.323, 0.837]
MiroThinker-1.7-mini	Unconstrained LLM	8/15	0.533	[0.266, 0.787]
MiroThinker-1.7-mini	EFR	12/15	0.800	[0.519, 0.957]

Note. PRC measures the proportion of intervention pairs ranked in the same order as the Pure-DCE reference. The policy set contains six interventions, yielding 15 unordered policy pairs. Brackets report exact binomial 95% confidence intervals. One-sided exact binomial tests were performed against the random-ranking baseline (PRC = 0.5). Benjamini–Hochberg correction was applied across the three EFR model comparisons (Qwen2.5-72B, DeepSeek V4 Pro, and MiroThinker-1.7-mini); Qwen2.5-72B and MiroThinker-1.7-mini remained significant after correction.

The results show that local deployment is feasible but that large local models impose a nontrivial inference cost. The Qwen2.5-72B local run required approximately 11.2 hours for 640 generations on an Apple M4 Max workstation, whereas the smaller local MiroThinker-1.7-mini model completed the same number of generations in approximately 4.2 minutes. The Static-BDT anchoring rule itself is computationally lightweight: once raw LLM probabilities and empirical frontier values are available, the post-generation correction requires only a convex probability-level calculation.

H Pure-DCE Reference and Policy-Ranking Validation

This appendix documents the Pure-DCE reference and the Policy Ranking Consistency (PRC) calculation reported in Table S9. The purpose is to make the policy-ranking exercise reproducible by specifying the policy set, eligible grid states, policy-effect calculation, pairwise ranking rule, and uncertainty interval construction.

H.1 Pure-DCE Reference

The Pure-DCE simulator uses the estimated empirical choice frontier directly:

$$\pi_{\text{DCE}}(y = 1 \mid \mathbf{x}) = P_{\text{static}}(y = 1 \mid \mathbf{x}).$$

It contains no LLM-generated component. In the PRC analysis, Pure-DCE serves as the ranking reference rather than as a competing LLM-based method. Its own PRC relative to itself is therefore mechanically equal to one and is not reported as a separate row in Table S9.

H.2 Candidate Policy Set

The PRC analysis uses six candidate interventions defined on the 64-state policy grid:

$$\mathcal{P} = \{p_1, p_2, p_3, p_4, p_5, p_6\},$$

where

$$\begin{aligned} p_1 &: \text{WaitTime } 6 \rightarrow 2, \\ p_2 &: \text{WaitTime } 4 \rightarrow 0, \\ p_3 &: \text{Efficacy } 0.50 \rightarrow 0.70, \\ p_4 &: \text{Efficacy } 0.70 \rightarrow 0.90, \\ p_5 &: \text{SideEffects } 3 \rightarrow 1, \\ p_6 &: \text{SideEffects } 2 \rightarrow 0. \end{aligned}$$

Each policy changes one target attribute while holding all non-target attributes fixed. For each intervention p , the eligible set $\mathcal{G}_p$ includes grid states whose baseline attribute equals the initial policy value and whose counterfactual state remains within the 64-state grid.

H.3 Policy-Level Average Effects

For model m , the average effect of policy p is computed as

$$\widehat{ATE}_p^m = \frac{1}{|\mathcal{G}_p|} \sum_{\mathbf{x}_g \in \mathcal{G}_p} [\pi_m(y = 1 \mid \mathbf{x}'_{g,p}) - \pi_m(y = 1 \mid \mathbf{x}_g)],$$

where $\mathbf{x}'_{g,p}$ is the counterfactual state obtained by applying policy p to baseline state $\mathbf{x}_g$. The evaluated probability functions are

$$\pi_m \in \{\pi_{\text{LLM}}, \pi_{\text{BDT}}, \pi_{\text{DCE}}\}.$$

Static-BDT probabilities are computed using the main anchoring specification:

$$\pi_{\text{BDT}} = 0.25 \bar{\pi}_{\text{LLM}} + 0.75 P_{\text{static}}.$$

H.4 Pairwise Ranking Rule

Let

$$\mathcal{P}_2 = \{(p, q) : p, q \in \mathcal{P}, p < q\}$$

denote the set of unordered policy pairs. With six candidate interventions, $|\mathcal{P}_2| = 15$. For each model m , PRC is computed as

$$\text{PRC}(m) = \frac{1}{|\mathcal{P}_2|} \sum_{(p,q) \in \mathcal{P}_2} \mathbf{1} \left[\text{sign}(\widehat{ATE}_p^m - \widehat{ATE}_q^m) = \text{sign}(\widehat{ATE}_p^{\text{DCE}} - \widehat{ATE}_q^{\text{DCE}}) \right].$$

Thus, PRC measures the share of policy-pair rankings that agree with the Pure-DCE reference. A value of one indicates that the model ranks all candidate interventions in the same order as the Pure-DCE simulator, while lower values indicate at least one pairwise reversal relative to the DCE-implied policy ranking.

H.5 Uncertainty Intervals

The confidence intervals reported in Table S9 are exact binomial 95% intervals over the 15 unordered policy-pair comparisons. For each model-method row, the number of correctly ranked policy pairs is treated as the number of successes out of 15 trials. These intervals summarize uncertainty over the finite set of policy-pair comparisons and should not be interpreted as respondent-level sampling uncertainty or as external validation against realized uptake.

H.6 Reproducibility Checks

The PRC computation is conducted on the same fixed 64-state grid used in the main frontier-alignment analysis. For each model, the analysis verifies that all 64 states are present, that the Pure-DCE probability equals P_{static} , and that the Static-BDT probability satisfies

$$\pi_{\text{BDT}} = 0.25 \bar{\pi}_{\text{LLM}} + 0.75 P_{\text{static}}$$

up to numerical tolerance. The resulting PRC values are reported in Table S9. The metric should be interpreted as policy-ranking consistency relative to the stated-preference frontier, not as external validation against realized vaccine uptake.

I Bootstrap Uncertainty Intervals

This appendix reports the uncertainty intervals used to qualify the main benchmark-relative results. The intervals are intended to summarize uncertainty over grid states, held-out observations, monotonicity-comparison pairs, and policy-pair rankings. They should not be interpreted as respondent-level sampling uncertainty from re-estimating the full 6-parameter conditional-logit frontier unless explicitly stated.

I.1 Bootstrap Procedure

To quantify uncertainty, we compute bootstrap confidence intervals tailored to the specific unit of analysis for each evaluation metric.

For **held-out prediction metrics** (Table S11), the DCE data possess a nested panel structure where multiple choice observations are clustered within individual respondents. To strictly account for within-respondent correlation and avoid underestimating standard errors, we employ a **respondent-level cluster bootstrap**. Specifically, we resample the 205 unique respondents (whose choices constitute the 809 matched observations) with replacement, maintaining the intact within-respondent choice structure. We perform 2,000 replications to ensure stable interval estimation. Notably, the resulting respondent-level confidence intervals are narrower than their observation-level counterparts. This counterintuitive but statistically sound result occurs because, in our specific data structure (205 clusters with approximately 4 observations each), structured cluster resampling provides a more stable variance estimate. It avoids the extreme duplication noise inherent in observation-level resampling, which can artificially inflate variance by breaking the within-respondent panel structure.

Table S10: Bootstrap intervals for selected frontier-alignment diagnostics.

Model	Method	Metric	Estimate	95% CI
Qwen2.5-72B	Unconstrained LLM	MSE	0.0258	[0.0198, 0.0331]
Qwen2.5-72B	Static-BDT Anchor	MSE	0.0016	[0.0012, 0.0021]
Qwen2.5-72B	Unconstrained LLM	MVR-Wait	0.6250	[0.5312, 0.7188]
Qwen2.5-72B	Static-BDT Anchor	MVR-Wait	0.3021	[0.1979, 0.4167]
Qwen2.5-72B	Unconstrained LLM	Spearman ρ	0.0608	[-0.1841, 0.2869]
Qwen2.5-72B	Static-BDT Anchor	Spearman ρ	0.6908	[0.5474, 0.8096]
MiroThinker-1.7-mini	Unconstrained LLM	MVR-SE	0.3333	[0.2396, 0.4271]
MiroThinker-1.7-mini	Static-BDT Anchor	MVR-SE	0.3333	[0.2396, 0.4271]

Note. Unlike the held-out prediction metrics in Table S11, which employ respondent-level cluster bootstrapping to account for the nested panel structure of DCE choices, the frontier-alignment metrics here are evaluated on the fixed 64-state policy grid. Therefore, frontier-deviation intervals are computed by resampling the 64 policy-grid states with replacement (1,000 replications), and MVR intervals are computed by resampling comparable attribute pairs. The MiroThinker-1.7-mini MVR-SE rows are retained to document a side-effect correction failure mode: with the corrected 6-parameter specification (Section ??) the side-effect coefficient has the smallest empirical signal among the five attributes, and the Static-BDT anchor at $\lambda = 0.25$ does not transmit a sufficient side-effect correction to move MiroThinker’s MVR-SE off the unconstrained value.

For **frontier-deviation metrics** (Table S10), the evaluation operates on the fixed 64-state policy grid rather than respondent-level choices. Therefore, we compute frontier-deviation intervals by resampling the 64 policy-grid states with replacement using 1,000 replications. For **monotonicity violation rates** (MVR), we resample comparable attribute pairs. For **policy-ranking consistency** (Table S12), we report exact binomial intervals over the 15 unordered policy-pair comparisons, as the evaluation is conducted over a finite set of policy pairs rather than a sampled distribution.

These intervals provide uncertainty summaries for the finite evaluation objects used in the paper: policy-grid states, held-out matched respondents, monotonicity-comparison pairs, and policy-pair rankings. They are not a substitute for external validation against realized uptake or for respondent-level uncertainty from re-estimating the full 6-parameter conditional-logit DCE frontier.

I.2 Frontier-Alignment Uncertainty

Table S10 reports bootstrap intervals for the primary frontier-alignment diagnostics discussed in the main text. The intervals support the main Qwen2.5-72B alignment result: the MSE and Spearman ρ intervals do not overlap between Static-BDT and the unconstrained LLM. The MVR-Wait intervals for Qwen2.5-72B also do not overlap (raw [0.531, 0.719], Static-BDT [0.198, 0.417]). The side-effect monotonicity diagnostic is weaker: for MiroThinker-1.7-mini, Static-BDT and the unconstrained LLM produce identical MVR-SE point estimates (0.333) and identical bootstrap CIs [0.240, 0.427], indicating that the Static-BDT anchor at $\lambda = 0.25$ does not transmit a sufficient side-effect signal on this frontier.

I.3 Held-Out Prediction Uncertainty

Table S11 reports bootstrap intervals for the held-out prediction metrics in Table S11. On the LLM-matched held-out subset, Static-BDT improves substantially over the unconstrained LLM for both

log loss and Brier score. The intervals for Static-BDT and the unconstrained LLM do not overlap for either metric. In contrast, the intervals for Static-BDT and Pure-DCE overlap for both log loss and Brier score, supporting the interpretation that Static-BDT brings raw LLM probabilities close to the structured DCE benchmark but should not be interpreted as outperforming Pure-DCE within the DCE attribute space.

Table S11: Bootstrap intervals for held-out DCE predictive performance.

Sample	Method	Metric	Estimate [95% CI]
Full held-out	Pure-DCE	Brier	0.208 [0.201, 0.214]
LLM-matched	Pure-DCE	Log loss	0.688 [0.676, 0.700]
LLM-matched	Unconstrained LLM	Log loss	0.815 [0.794, 0.836]
LLM-matched	Static-BDT Anchor	Log loss	0.703 [0.692, 0.714]
LLM-matched	Pure-DCE	Brier	0.247 [0.241, 0.252]
LLM-matched	Unconstrained LLM	Brier	0.308 [0.298, 0.318]
LLM-matched	Static-BDT Anchor	Brier	0.255 [0.249, 0.260]

Note. Held-out intervals are computed via **respondent-level cluster bootstrapping** (2,000 replications, resampling the 205 unique respondents with replacement) to account for within-respondent correlation in the panel DCE data. Log-loss intervals are reported for the LLM-matched subset, where Pure-DCE, the unconstrained LLM, and Static-BDT can be compared on the same observations. The full held-out row reports the Brier interval for Pure-DCE on all held-out observations.

I.4 Policy-Ranking Consistency Intervals

Table S12 reports exact binomial 95% intervals for Policy Ranking Consistency (PRC), calculated over the 15 unordered policy-pair comparisons. For Qwen2.5-72B, the point estimate increased from 0.333 for the unconstrained LLM to 0.867 after anchoring. The corresponding intervals are wide because the policy-ranking exercise contains only 15 pairwise comparisons. These intervals therefore summarize uncertainty over the finite policy set and should not be interpreted as respondent-level uncertainty or external validation of policy rankings.

J Covariate Balance Check: Matched vs. Unmatched Subset

To assess the representativeness of the 21.9% matched held-out subset ($N = 809$) used in Table 2, we compare its covariate distribution against the unmatched subset ($N = 2,881$). Table S13 reports the means and Standardized Mean Differences (SMD) for key DCE attributes and the choice rate.

The results confirm that the matched subset systematically differs from the unmatched subset on all observed dimensions (all SMDs > 0.1). The root cause is structural rather than sampling-based: the 64-state LLM grid evaluates only specific integer attribute levels, inherently oversampling longer waits, higher efficacy, and higher side-effect burdens. Consequently, the matched subset serves as a harder, non-representative test bed for the specific attribute profiles covered by the grid. Importantly, because all 205 respondents are present in both subsets, this covariate imbalance does not compromise the respondent-level cluster bootstrap intervals reported in the main text.

Table S12: Exact binomial intervals for Policy Ranking Consistency.

Model	Method	Matching pairs	PRC	95% CI
Qwen2.5-72B	Unconstrained LLM	5/15	0.333	[0.118, 0.616]
Qwen2.5-72B	Static-BDT Anchor	13/15	0.867	[0.595, 0.983]
DeepSeek V4 Pro	Unconstrained LLM	2/15	0.133	[0.017, 0.405]
DeepSeek V4 Pro	Static-BDT Anchor	9/15	0.600	[0.323, 0.837]
MiroThinker-1.7-mini	Unconstrained LLM	8/15	0.533	[0.266, 0.787]
MiroThinker-1.7-mini	Static-BDT Anchor	12/15	0.800	[0.519, 0.957]

Note. PRC intervals are exact binomial 95% intervals over the 15 unordered policy-pair comparisons. They summarize uncertainty over the finite policy-pair ranking exercise and should not be interpreted as respondent-level uncertainty or external policy validation.

Table S13: Covariate balance between matched and unmatched held-out subsets.

Variable	Matched Mean	Unmatched Mean	SMD
Wait Time (months)	2.77	1.28	0.61
Vaccine Efficacy	0.59	0.45	0.49
Side Effects	1.94	1.17	0.76
Cash Incentives (RMB)	331	206	0.40
Choice Rate ($y = 1$)	0.519	0.281	0.50

Note. The matched subset consists of held-out observations whose attribute profiles exactly match the 64-state LLM evaluation grid. SMD denotes the Standardized Mean Difference. The systematic differences arise because the 64-state grid restricts attributes to specific integer levels (e.g., WaitTime $\in \{0, 2, 4, 6\}$), thereby oversampling more extreme profiles relative to the full DCE design which includes intermediate levels. Crucially, all 205 held-out respondents contribute to both subsets; thus, the imbalance is strictly at the observation (attribute-level) rather than the respondent level.

K Subgroup Heterogeneity Analysis

To assess whether the hybrid framework’s performance is uniform across respondent subgroups, we replicated the held-out predictive validation on the 809-respondent LLM-matched subset stratified by age, gender, and education. Table S14 reports the log loss, Brier score, and Spearman rank correlation of the Static-BDT anchored predictions within each subgroup, alongside the unconstrained Qwen2.5-72B baseline for direct comparison. Because the 809-respondent subset is too small for stable within-subgroup bootstrap inference, the table reports point estimates only; bootstrap intervals on the full sample are reported in the main text.

Two patterns emerge. First, the unconstrained LLM produces near-zero or slightly negative Spearman correlation with the empirical frontier in every subgroup, confirming that the behavioral unreliability documented in the main text is not driven by a particular demographic segment. Second, the Static-BDT Spearman correlation falls in a narrow band of 0.948–0.955 across all seven subgroups (range 0.007), with no subgroup falling below the 0.94 threshold. This near-uniform behavior is consistent with the Static-BDT anchoring rule being a global probability-blending operation that does not condition on respondent characteristics; the empirical frontier itself is population-averaged by construction, and the resulting anchoring benefit is therefore expected to transfer across subgroups defined by age, gender, and education within this sample.

These subgroup results should not be over-interpreted. The 809-respondent matched subset does not provide sufficient statistical power to detect small subgroup-specific effects, and the Wuhan

Table S14: Subgroup heterogeneity analysis of Static-BDT ($\lambda = 0.25$) performance on the 809-respondent LLM-matched held-out subset, stratified by demographic characteristics.

Subgroup	Method	Log loss ↓	Brier ↓	Spearman ρ ↑
Age 18–44	Unconstrained LLM	0.812	0.305	−0.04
	Static-BDT	0.701	0.253	0.953
Age 45–65	Unconstrained LLM	0.820	0.312	−0.01
	Static-BDT	0.706	0.258	0.951
Age 65+	Unconstrained LLM	0.819	0.310	−0.05
	Static-BDT	0.705	0.257	0.949
Female	Unconstrained LLM	0.813	0.307	−0.03
	Static-BDT	0.702	0.254	0.951
Male	Unconstrained LLM	0.817	0.309	−0.02
	Static-BDT	0.704	0.256	0.953
Lower education	Unconstrained LLM	0.818	0.310	−0.04
	Static-BDT	0.706	0.258	0.948
Higher education	Unconstrained LLM	0.811	0.305	−0.03
	Static-BDT	0.700	0.253	0.955

Note. Lower education corresponds to middle school or below; higher education corresponds to high school, college, or graduate degree. The unconstrained LLM column reports Qwen2.5-72B predictions before anchoring. Across all seven subgroups, the Static-BDT predictions achieve Spearman ρ between 0.948 and 0.955, a range of 0.007, indicating that the empirical anchoring benefit is essentially invariant to demographic subgroup within this Wuhan sample.

sample is not designed to be demographically representative of any specific target population. The subgroup analysis is presented as a robustness check on the homogeneity of the anchoring effect, not as evidence for the generalizability of the empirical frontier itself. Future work with larger and more diverse samples is required to assess whether the framework’s anchoring benefit is preserved across demographic groups in other populations.

L Replication Materials

Replication materials are available in a public GitHub repository: <https://github.com/yichao2022/behavioral-digital-twins>. The repository contains code and supporting files used to reproduce the main BDT analyses, including prompt templates, grid-level model outputs, empirical-frontier probabilities, Static-BDT anchoring scripts, metric-computation code, bootstrap-interval scripts, and scripts for reproducing the main tables and figures.

Individual-level DCE responses are not publicly released because they involve human-subject survey data. To support reproducibility while preserving respondent confidentiality, the repository provides de-identified grid-level inputs and model-level outputs required to reproduce the frontier-alignment, held-out validation, out-of-design stress-test, policy-ranking, lambda-sensitivity, bootstrap-interval, and runtime analyses.

References

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